

The periodic spin-one Haldane gap

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Abstract

We prove a uniform positive spectral gap for the pure antiferromagnetic spin-one Heisenberg chain on even periodic rings, establishing the positive even-periodic formulation of the spin-one Haldane conjecture.

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1 Introduction

Haldane predicted that antiferromagnetic Heisenberg chains have different low-energy behavior for integer and half-odd-integer spins. The prediction was first formulated in 1981 and published in 1983 [13, 15, 14]. For spin one, the prediction is a positive gap above the ground energy. We prove this prediction for the pure bilinear model on even periodic rings, with explicit bounds uniform in the length.

Let $S^x, S^y, S^z \in M_3(\mathbb{C})$ be the self-adjoint irreducible spin-one matrices, normalized by

$$[S^\alpha, S^\beta] = i \sum_{\eta} \varepsilon_{\alpha\beta\eta} S^\eta, \quad \sum_{\alpha} (S^\alpha)^2 = 2I_3.$$

For each even integer $L \geq 4$, let $\mathcal{H}_L = (\mathbb{C}^3)^{\otimes L}$ and define

$$H_L = \sum_{j=0}^{L-1} \sum_{\alpha=x,y,z} S_j^\alpha S_{j+1}^\alpha, \quad S_L^\alpha = S_0^\alpha. \quad (1.1)$$

Here S_j^α acts on site j and is the identity on the other factors. The coupling constant is one. Write $E_0(L)$ for the least energy and $E_1(L)$ for the next *distinct* energy of H_L , and set

$$\gamma_L = E_1(L) - E_0(L), \quad \Delta_1 = \liminf_{\substack{L \rightarrow \infty \\ L \text{ even}}} \gamma_L. \quad (1.2)$$

These definitions concern the full spectrum, including the multiplicity of the ground energy. The next distinct energy exists: the all-zero and all-+1 product vectors in the S^z basis have energy expectations 0 and L , respectively, so H_L is not scalar.

Theorem 1.1. *For the Hamiltonians (1.1), the ground state is unique for every even $L \geq 60$, and*

$$\gamma_L > \frac{4}{105} \log \frac{80}{79} \quad \text{for every even } L \geq 60, \quad (1.3)$$

$$\gamma_L > \frac{\log 20}{784} \quad \text{for every even } L \geq 2304. \quad (1.4)$$

Consequently

$$\Delta_1 \geq \frac{\log 20}{784} > 0.$$

The spectral conclusions also hold for any simultaneously similar spin-one realization satisfying the same commutator and Casimir normalizations.

Theorem 1.1 resolves the spin-one Haldane gap conjecture positively in the even-periodic finite-volume formulation $\Delta_1 > 0$. It also implies $\inf_{L \geq 4, L \text{ even}} \gamma_L > 0$: each of the finitely many remaining finite-dimensional Hamiltonians has a positive gap above its full ground sector. This observation supplies neither a numerical value nor a simplicity assertion at those small lengths. Corollary 6.1 also gives the corresponding lower bound for local excitations in every subsequential thermodynamic limit of the periodic ground states of the self-adjoint model.

1.1 Historical context

The half-odd-integer side has a rigorous antecedent in the low-energy excitation bound of Lieb, Schultz and Mattis for the cyclic spin-one-half chain [17, Appendix B, Theorem 2]. Affleck and Lieb extended this obstruction to the half-odd-integer antiferromagnetic chains they studied [3]. The

possible alternative of degenerate ground states is part of that obstruction. Integer spin permits a unique gapped state, but this distinction alone does not prove a gap for a particular interaction.

The Affleck–Kennedy–Lieb–Tasaki construction established a rigorous gapped spin-one chain with a biquadratic interaction [1, 2]. Its bond interaction is $h + \frac{1}{3}h^2$, up to a scalar and normalization, where $h = \mathbf{S}_j \cdot \mathbf{S}_{j+1}$. The Hamiltonian in (1.1) has bond interaction h itself. Yarotsky proved that a unique ground state and a uniform periodic gap persist under sufficiently small translation-invariant finite-range perturbations of the AKLT interaction [24, arXiv v1, Theorems 1 and 3]. This controls a neighborhood of that interaction; it does not establish a gapped path to the pure bilinear point.

For the pure spin-one chain, White and Huse’s density-matrix renormalization-group study estimated a bulk gap of 0.41050(2) and exhibited effective spin-one-half degrees of freedom at open ends [23]. Neutron scattering provided experimental evidence in the nearly one-dimensional spin-one antiferromagnet CsNiCl_3 [8]. These results explain both the strong physical evidence for the prediction and the significance of boundary conditions. The uniform lower bounds below instead follow from the stated finite certificates and an analytic argument valid at every larger even length.

Batista and Ortiz proposed a spin–fermion proof of the Haldane gap [6]. Their gap argument following Equation (8) of the preprint version uses a first-order perturbative identification with an XYZ chain; that step does not supply a uniform remainder estimate or a continuation to the original pairing coefficient. More recently, Tasaki established a nontrivial infinite-chain topological index for the spin-one Heisenberg model under a gap assumption for odd open chains with endpoint fields [22, arXiv v4, Assumption 2 and Theorem 3]. The boundary hypotheses of that theorem require a separate argument from the even-periodic estimates proved here.

Finite-volume methods for uniform spectral gaps have a substantial history, including the criteria of Knabe and Nachtergaele for frustration-free chains, whose ground states minimize every local interaction [16, 18]. Gosset and Mozgunov obtained sharper finite-size thresholds within that frustration-free setting [12]. Recently, Banerjee, Guo and Chowdhury developed semidefinite bounds for string correlators using an enclosing ground-energy window, including finite periodic spin-one Heisenberg chains [5, Equations (2), (13) and Figure 4]. Those observable bounds concern a different quantity from the uniform spectral gap studied here. Our finite inputs are thermal traces of small twisted chains and variational energy estimates from periodic matrix-product vectors, in the tradition of finitely correlated states [9]. The latter use the standard trace-of-matrix-products representation [20, Section 2.1]; we give the particular integer matrices and all contractions explicitly.

The finite thermal inputs are rigorous enclosures of matrix-exponential traces. Validated matrix exponentiation has a long history, including self-validating Padé methods [7] and interval polynomial evaluation [11]. Our particular implementation uses a conditioned Poisson polynomial in a contraction. The factor-two conditioning estimate follows the coefficient argument of Fox and Glynn [10, Proposition 1]. Explicit bounds for the integer evaluation and the final trace enclosure are supplied in the appendices.

1.2 How the proof works

Quantum transfer-matrix methods relate partition functions to spatial moments [21]. Our construction uses an ordered bond expansion of the type developed by Aizenman and Nachtergaele [4, arXiv v1, Section 2.2], followed by a direct operator construction on bond histories.

A shifted partition function is a sum of positive physical Boltzmann weights. We construct a self-adjoint spatial transfer operator whose L th moment is the same partition function. For even L , the spatial moment is also a sum of nonnegative weights. This gives two probability distributions associated with one partition function. The *purity* of a distribution (p_i) is $\sum_i p_i^2$; purity close to one means that a single weight carries almost all of its mass.

Squaring and renormalizing a distribution reduces its purity defect quadratically once the defect is small. Squaring physical weights doubles inverse temperature, while squaring spatial weights doubles length. Exact cancellation identities couple these two improvements. Starting from two finite purity bounds, both defects then decay doubly exponentially as length and inverse temperature double together. Spatial moment inequalities cover every intermediate even length. Finally, purity of the full physical Gibbs distribution bounds each excited-to-ground probability ratio and hence the physical energy gap. Proposition 4.3 states this coupled argument as a general gap criterion. Its assumptions isolate a reusable criterion for other Hamiltonian families with the same two partition-function representations; positivity of the spatial transfer is not required.

There are two finite initializations. One starts the iteration at length 60 and inverse temperature 105/4. The other reaches length 2304 and inverse temperature 784 with a smaller purity defect, giving the stronger eventual bound. They share the transfer construction and the bootstrap, while using different thermal moments and two lengths of one matrix-product trial family. The finite computations involve integer arithmetic, rational interval bounds and explicit truncation errors; no floating-point eigenvalue estimate is a premise.

Section 2 fixes the model and proves its elementary energy bounds. Section 3 constructs the spatial transfer and its rotation sectors. Section 4 proves the common purity theorem, and Section 5 establishes the two finite initializations from explicitly stated thermal and trial inputs. Section 6 combines these results to prove Theorem 1.1. Appendices A–C prove the essential thermal and trial inputs, with exact data and reproducible finite procedures.

2 The Hamiltonian and its auxiliary twists

The argument uses the same partition function in two different ways: as a Gibbs trace in the physical Hilbert space and as a moment of a spatial operator. We first choose coordinates that make both descriptions explicit. We also establish an elementary energy bound used to control the finite exponential approximations.

2.1 Spin coordinates

We use the self-adjoint spin-one matrices. This convention does not restrict the spectrum in the formulation where self-adjointness is not assumed.

Lemma 2.1 (Choice of representation). *Let $S^x, S^y, S^z \in M_3(\mathbb{C})$ satisfy*

$$[S^\alpha, S^\beta] = i \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} S^\gamma, \quad \sum_{\alpha} (S^\alpha)^2 = 2I.$$

They are simultaneously similar to the standard self-adjoint spin-one matrices. If the original matrices are self-adjoint, the similarity can be chosen unitary. The corresponding tensor-power similarity preserves the complete spectrum of every periodic Hamiltonian H_n .

Proof. Put $S^\pm = S^x \pm iS^y$. The assumptions give

$$[S^z, S^\pm] = \pm S^\pm, \quad S^- S^+ = 2I - (S^z)^2 - S^z, \quad S^+ S^- = 2I - (S^z)^2 + S^z. \quad (2.1)$$

Choose an S^z eigenvector v whose eigenvalue m has maximal real part. Then $S^+ v = 0$, since otherwise it has eigenvalue $m + 1$. Thus $m = 1$ or $m = -2$. The latter case would permit indefinite nonzero lowering: a last nonzero lowered vector would have weight $-2 - k$, whereas $S^+ S^- w = 0$ forces its weight to be -1 or 2 . This contradicts finite dimension. Hence $m = 1$.

The vectors

$$v, \quad \frac{S^-v}{\sqrt{2}}, \quad \frac{(S^-)^2v}{2}$$

are nonzero by (2.1) and have distinct weights $1, 0, -1$. They form a basis. There is no further lowered vector, because the basis has no weight -2 . The identities give raising and lowering coefficients $\sqrt{2}$, so this is the standard spin-one representation. If the original matrices are self-adjoint and $\|v\| = 1$, the three vectors are orthonormal. Finally, conjugating each tensor factor conjugates every term of H_n . $\square$

In the weight basis $|s\rangle$, $s \in \{-1, 0, 1\}$, the diagonal generator is $S^z|s\rangle = s|s\rangle$ and the nonzero ladder entries are $\sqrt{2}$. We will also use the Cartesian basis, indexed by the three axes, in which

$$(S^\alpha)_{ij} = -i\varepsilon_{\alpha ij}, \quad G_\alpha = iS^\alpha.$$

These matrices satisfy the same relations. Each G_α is real and has operator norm one. For the two-site interaction $h = \sum_\alpha S^\alpha \otimes S^\alpha$, contraction of the epsilon tensors gives

$$-h = \sum_\alpha G_\alpha \otimes G_\alpha, \quad h = F - |\Omega\rangle\langle\Omega|, \quad \Omega = \sum_{i=1}^3 |ii\rangle, \quad (2.2)$$

where $F(u \otimes v) = v \otimes u$. In particular, h has eigenvalues $-2, -1, 1$, and $h \leq I$.

2.2 Twists and partition functions

Number the sites $0, \dots, n-1$, with $n \geq 4$. A proper Cartesian rotation $g \in SO(3)$ is unitary on $\mathbb{C}^3$. Define

$$H_n(g) = \sum_{j=0}^{n-2} h_{j,j+1} + h_{n-1,0}(g), \quad h_{n-1,0}(g) = \sum_\alpha S_{n-1}^\alpha (g S^\alpha g^{-1})_0. \quad (2.3)$$

Thus only the last bond is conjugated, on its site-0 factor. The Hamiltonian is self-adjoint, and $H_n(I) = H_n$. Fix

$$a = \frac{700741}{500000}, \quad Z_n(b, g) = \text{Tr} \exp[-b(H_n(g) + anI)], \quad Z_n(b) = Z_n(b, I) \quad (b > 0). \quad (2.4)$$

All these partition functions are strictly positive. The scalar shift will cancel from the purity ratios and from the final spectral gap. With this normalization, the trial bounds in Lemma C.2 give $E_0(n) + an < 0$ at $n = 72, 120$, and hence $Z_n(b) > 1$ for every $b > 0$. These lower bounds are paired with finite thermal upper bounds in Section 5.

We need the diagonal rotations

$$P_x = \text{diag}(1, -1, -1), \quad P_y = \text{diag}(-1, 1, -1), \quad P_z = \text{diag}(-1, -1, 1), \quad D_2 = \{I, P_x, P_y, P_z\},$$

and the cyclic rotation $C : e_x \mapsto e_y \mapsto e_z \mapsto e_x$. Write $P = P_x$. The three nonidentity elements of D_2 have equal partition functions, by a global cyclic rotation of all sites. The rotation C has angle $2\pi/3$ about the diagonal axis. After a global rotation and a change to the weight basis, P and C are respectively represented by $\exp(-i\theta S^z)$ with $\theta = \pi$ and $\theta = 2\pi/3$.

For later comparison with finite matrices, a word $x = (x_0, \dots, x_{n-1})$ in the weight basis has diagonal entry $\sum_j x_{j-1}x_j$, with indices modulo n . Each permitted step changing two neighboring letters by opposite units has coefficient one. For a seam step from column y to row x , this coefficient is

$$\exp(i\theta(y_0 - x_0)). \quad (2.5)$$

Indeed $gS^\pm g^{-1} = e^{\mp i\theta} S^\pm$. These formulas identify the finite matrices used below with the physical Hamiltonians in (2.3).

2.3 A uniform energy bound

Lemma 2.2 (Energy bounds). *For every $n \geq 4$ and every $g \in SO(3)$,*

$$-\frac{3n}{2}I \leq H_n(g) \leq nI. \quad (2.6)$$

Proof. The upper bound follows from $h \leq I$ and unitary conjugation of the seam. For the lower bound, consider $T = -h_{12} - h_{23}$ on three open sites in Cartesian coordinates. By (2.2), an unequal pair swaps with coefficient -1 , and an equal pair changes to either of the other equal pairs with coefficient 1 . There are no diagonal terms.

Assign a positive weight q to each three-letter word. The following table lists every equality pattern; a, b, c denote distinct letters. The last column sums the weights of the neighbors of a row word, with the absolute matrix coefficients.

word type	q	absolute weighted row sum
aaa	4	$3 + 3 + 3 + 3 = 12$
aab or abb	3	$4 + 3 + 2 = 9$
aba	2	$3 + 3 = 6$
abc	2	$2 + 2 = 4$

Every row sum is at most $3q$. If D is the diagonal matrix of these weights, $D^{-1}TD$ has absolute row sums at most three. Its spectral radius is therefore at most three. Since T is self-adjoint and similar to this matrix, $h_{12} + h_{23} \geq -3I$.

An open triad containing the seam is unitarily equivalent to an untwisted triad: delete the seam bond and rotate all sites in one of the two resulting components. The internal interaction of that component is rotation invariant. Thus the same lower bound holds for every consecutive pair of bonds in $H_n(g)$. Summing the n triad inequalities counts each bond twice and proves (2.6). $\square$

3 Partition functions as spatial moments

We now construct a self-adjoint spatial operator whose moments are the partition functions (2.4). This identity lets even spatial powers play the same role as positive Gibbs weights. The operator itself need not be positive.

Let $\mathcal{G} \subset SO(3)$ be the group of proper signed permutation matrices. It contains D_2 and C . For $g \in \mathcal{G}$, write

$$ge_\alpha = \epsilon_g(\alpha)e_{\sigma_g(\alpha)}, \quad \epsilon_g(\alpha) \in \{-1, 1\}.$$

Rotation covariance of the Cartesian matrices gives

$$gG_\alpha g^{-1} = \epsilon_g(\alpha)G_{\sigma_g(\alpha)}. \quad (3.1)$$

Proposition 3.1 (Spatial transfer). *For each $b > 0$ there are a Hilbert space $\mathcal{K}_b$, a compact self-adjoint operator X_b , and a unitary representation $g \mapsto U_g$ of $\mathcal{G}$ commuting with X_b , such that*

$$Z_n(b, g) = \text{Tr}(X_b^n U_g) \quad (n \geq 4, g \in \mathcal{G}). \quad (3.2)$$

The operator X_b and the representation commute with complex conjugation. Moreover X_b is Hilbert-Schmidt, so X_b^k is trace class for every integer $k \geq 2$. If $\lambda_i(b)$ are its nonzero eigenvalues, counted with multiplicity, then

$$\sum_i |\lambda_i(b)|^k < \infty \quad (k \geq 2), \quad Z_n(b) = \sum_i \lambda_i(b)^n \quad (n \geq 4). \quad (3.3)$$

Proof. Histories and the kernel. A bond history is a finite ordered list

$$x = ((t_1, \alpha_1), \dots, (t_k, \alpha_k)), \quad 0 < t_1 < \dots < t_k < b, \quad \alpha_j \in \{x, y, z\},$$

including the empty list when $k = 0$. Let Ω_b be the disjoint union of these labeled simplexes. Give each simplex Lebesgue measure and the empty history mass one. The resulting measure μ is finite:

$$\mu(\Omega_b) = \sum_{k=0}^{\infty} \frac{3^k b^k}{k!} = e^{3b}. \quad (3.4)$$

Set $\mathcal{K}_b = L^2(\Omega_b, \mu; \mathbb{C})$.

For two histories x, y , merge their labeled times in increasing order and define $k_b(x, y)$ as the trace of the resulting product of the matrices G_α , with increasing time from left to right. A nonempty time collision is a null event; set the kernel to zero there. The product for two empty histories is the identity, so its trace is three.

The kernel is real and has absolute value at most three, since $\|G_\alpha\| = 1$. Exchanging x and y leaves their merged chronological list unchanged. Hence $k_b(x, y) = k_b(y, x)$; no reversal of the matrix product occurs. Its integral operator K_b is therefore self-adjoint and Hilbert–Schmidt, with

$$\|K_b\|_{\text{HS}}^2 \leq 9\mu(\Omega_b)^2 = 9e^{6b}.$$

Define $X_b = e^{-ab} K_b$. Compact self-adjoint spectral theory gives real, square-summable eigenvalues. Since they are bounded, all their absolute moments of integer order at least two are summable. This proves the operator assertions about X_b .

Rotations. The permutation σ_g acts on the labels of a history, preserving its times and the measure. Put

$$s_g(x) = \prod_{\alpha \text{ in } x} \epsilon_g(\alpha), \quad (U_g f)(x) = s_g(\sigma_g^{-1} x) f(\sigma_g^{-1} x).$$

The rule $s_{gh}(x) = s_g(\sigma_h x) s_h(x)$ proves that these real unitaries form a representation. Simultaneously rotating every matrix inside a site trace, using (3.1), gives

$$k_b(\sigma_g x, \sigma_g y) = s_g(x) s_g(y) k_b(x, y).$$

It follows that U_g commutes with K_b and X_b . In particular the kernel of $K_b U_g$ is

$$(K_b U_g)(x, y) = k_b(x, \sigma_g y) s_g(y). \quad (3.5)$$

The cycle trace. For $n \geq 2$, both K_b^{n-1} and $K_b U_g$ are Hilbert–Schmidt. The kernel of K_b^{n-1} is the iterated kernel obtained by integration; it is bounded because k_b is bounded and μ is finite. The Hilbert–Schmidt product trace identity

$$\text{Tr}(AB) = \int a(x, y) b(y, x) d\mu(x) d\mu(y)$$

and (3.5) consequently give

$$\text{Tr}(K_b^n U_g) = \int_{\Omega_b^n} \left(\prod_{j=0}^{n-2} k_b(x_j, x_{j+1}) \right) k_b(x_{n-1}, \sigma_g x_0) s_g(x_0) d\mu^n. \quad (3.6)$$

The product trace identity follows from the Hilbert–Schmidt inner product formula and Cauchy–Schwarz. All the iterated kernel integrals here are absolutely integrable. Thus (3.6) does not require a diagonal-kernel formula for K_b itself.

Identification with the physical partition function. The ordered bond expansion is a standard representation of spin partition functions [4, arXiv v1, Section 2.2, Equations (2.10)–(2.12)]. We verify it here with the required seam and normalization. Expand $\exp[-bH_n(g)]$ in its ordered-time series. By (2.2), each event chooses a bond and one of three labels, and inserts the corresponding two G factors. At the seam the site-0 factor is rotated as in (3.1). There are $(3n)^k$ choices at total degree k ; each tensor operator has norm at most one. The sum of the absolute traced contributions is bounded by

$$3^n \sum_{k=0}^{\infty} \frac{(3nb)^k}{k!} = 3^n e^{3nb}. \quad (3.7)$$

We may therefore regroup all terms and integrals by bond. The interleavings of the bond histories partition the global time simplex up to null collisions. At each site, the tensor trace is the kernel on the two incident histories, in their original chronological order.

For an explicit check of the seam, call its history e_0 and enumerate the remaining bond histories $e_1, \dots, e_{n-1}$ around the ring. The physical integrand is

$$s_g(e_0)k_b(\sigma_g e_0, e_1)k_b(e_1, e_2) \cdots k_b(e_{n-1}, e_0).$$

In (3.6), set $x_0 = e_0, x_1 = e_{n-1}, \dots, x_{n-1} = e_1$ and use kernel symmetry. This gives exactly the displayed integrand, with g on site 0. The spatial enumeration has been reversed; the chronological products have not. Thus $\text{Tr} e^{-bH_n(g)} = \text{Tr}(K_b^n U_g)$. Multiplication by e^{-abn} proves (3.2). The ordinary trace formula for a compact self-adjoint operator now gives (3.3). $\square$

For an even $n \geq 4$, Proposition 3.1 supplies the probabilities

$$p_i = \frac{|\lambda_i(b)|^n}{Z_n(b)}, \quad \sum_i p_i = 1, \quad \sum_i p_i^2 = \frac{Z_{2n}(b)}{Z_n(b)^2}. \quad (3.8)$$

Odd moments will also be useful in polynomial estimates, but evenness is essential in (3.8). These spatial eigenvalues are distinct objects from the physical energies of H_n .

3.1 Rotation sectors

The small twisted partition functions separate the spatial eigenvalues into a few lists. Their multiplicities will help isolate a dominant weight; no restriction on the physical spectrum is imposed.

Lemma 3.2 (Sector moments). *Fix $b > 0$ and let X_b, U_g be as in Proposition 3.1. There are real eigenvalue lists, counted with multiplicity, whose integer moments I_k, O_k, N_k converge absolutely for $k \geq 2$, with*

$$Z_n(b) = I_n + 2O_n + 3N_n, \quad (3.9)$$

$$Z_n(b, P) = I_n + 2O_n - N_n, \quad (3.10)$$

$$Z_n(b, C) = I_n - O_n \quad (n \geq 4). \quad (3.11)$$

The I list is the restriction to the joint invariant subspace of D_2 and C . The O list occurs twice and the N list three times in the full eigenvalue list of X_b . All three moments are nonnegative at even orders. If J_n denotes the moment on the D_2 -invariant subspace, then

$$J_n = I_n + 2O_n.$$

Proof. The four characters χ of D_2 define orthogonal projections

$$\Pi_\chi = \frac{1}{4} \sum_{g \in D_2} \chi(g) U_g.$$

They commute with X_b and sum to the identity. Conjugation by U_C permutes the three nontrivial character spaces cyclically. The restrictions of X_b to these spaces are therefore unitarily equivalent; denote their common moment by N_k .

In the trivial-character space, split further according to the eigenvalues $1, \omega, \bar{\omega}$ of U_C , where $\omega = e^{2\pi i/3}$. These are reducing subspaces for X_b . Complex conjugation exchanges the last two spaces and preserves every real eigenvalue of X_b , with multiplicity. Their common moment is O_k ; the moment on the first space is I_k . Absolute convergence follows from (3.3).

This orthogonal decomposition proves (3.9). On the three nontrivial character spaces, a fixed U_P has signs $+1, -1, -1$, while it is the identity on the trivial-character space. This gives (3.10). The operator U_C permutes the nontrivial spaces, so they contribute zero to $\text{Tr}(X_b^n U_C)$. On the remaining three spaces its coefficients are $1, \omega, \bar{\omega}$, and $\omega + \bar{\omega} = -1$. This proves (3.11). The identity for J_n follows from the decomposition of the trivial-character space; even powers give nonnegativity. $\square$

The invariant I space need not be one-dimensional. The point of Lemma 3.2 is that every weight outside it occurs at least twice in the full list. Later moment bounds will identify a unique largest weight without discarding any sector.

4 From two purities to a uniform gap

The spatial and physical descriptions of the same partition function give two ways to concentrate its weights. This section isolates the argument that propagates two initial concentration estimates to all larger even lengths. The initial estimates are established in Section 5.

By Proposition 3.1, each partition function is both a physical Gibbs trace and, at even lengths, a nonnegative spatial moment. The former counts every physical eigenvector, with multiplicity; the latter counts the absolute powers of every spatial eigenvalue. These two representations give the two probability distributions used below.

For even $n \geq 4$, define the spatial and physical purities

$$S(n, b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \quad T(n, b) = \frac{Z_n(2b)}{Z_n(b)^2}. \quad (4.1)$$

The energy shift aL cancels from both ratios. We first record the elementary inequality responsible for their improvement under squaring.

Lemma 4.1 (Squaring probabilities). *Let (w_i) be a finite or countable family of nonnegative numbers with $\sum_i w_i = 1$. Its purity $P = \sum_i w_i^2$ lies in $(0, 1]$. The normalized squares w_i^2/P have purity P' satisfying*

$$1 - P' \leq f(1 - P), \quad f(u) = \frac{u^2}{2(1 - u)^2} \quad (0 \leq u < 1). \quad (4.2)$$

The function f is increasing. Consequently

$$\begin{aligned} 0 < S(n, b), T(n, b) &\leq 1, \\ 1 - S(2n, b) &\leq f(1 - S(n, b)), \\ 1 - T(n, 2b) &\leq f(1 - T(n, b)). \end{aligned} \quad (4.3)$$

Proof. At least one weight is positive, so $P > 0$, and $w_i^2 \leq w_i$ gives $P \leq 1$. Nonnegativity permits rearrangement of all countable sums. Thus

$$1 - P' = \frac{2 \sum_{i < j} w_i^2 w_j^2}{P^2} \leq \frac{2 (\sum_{i < j} w_i w_j)^2}{P^2} = \frac{(1 - P)^2}{2P^2}.$$

Also $f'(u) = u/(1 - u)^3 \geq 0$.

For $S(n, b)$, use the spatial probabilities $|\lambda_i(b)|^n / Z_n(b)$ from (3.8). Squaring them changes n to $2n$. For $T(n, b)$, list all physical eigenvalues $\varepsilon_k(n)$ with multiplicity and use

$$w_k(n, b) = \frac{e^{-b(\varepsilon_k(n) + an)}}{Z_n(b)}. \quad (4.4)$$

Their purity is $T(n, b)$, and squaring them changes b to $2b$. This proves (4.3). $\square$

4.1 The coupled update

The two purities constrain one another because their definitions use the same partition functions. Direct cancellation gives

$$S(n, 2b) = S(n, b)^2 \frac{T(2n, b)}{T(n, b)^2}, \quad (4.5)$$

$$T(4n, b) = T(2n, b)^2 \frac{S(2n, 2b)}{S(2n, b)^2}. \quad (4.6)$$

Both denominators are at most one. The following form allows the initial bounds to have different sizes and to be rounded upward.

Lemma 4.2 (Coupled update). *Suppose $n \geq 4$ is even, $b > 0$, and $p, q \in [0, 1)$ satisfy*

$$1 - S(n, b) \leq p, \quad 1 - T(2n, b) \leq q.$$

If $p_+, q_+ \in [0, 1)$ obey

$$\begin{aligned} p_+ &\geq f(1 - (1 - p)^2(1 - q)), \\ q_+ &\geq f(1 - (1 - q)^2(1 - p_+)), \end{aligned} \quad (4.7)$$

then

$$1 - S(2n, 2b) \leq p_+, \quad 1 - T(4n, 2b) \leq q_+.$$

Proof. Equation (4.5) gives $S(n, 2b) \geq (1 - p)^2(1 - q)$. Applying spatial squaring in Lemma 4.1 and monotonicity of f proves the first bound. Next (4.6) gives $T(4n, b) \geq (1 - q)^2(1 - p_+)$. Temporal squaring proves the second. Every input of f is in $[0, 1)$ because the product subtracted from one is positive and at most one. $\square$

In particular, one may round the first output upward before using it in the second update, then round the second output upward. The pair based at (n, b) becomes a pair based at $(2n, 2b)$: its physical member is at length $4n$. No decrease of the individual bounds is needed for this step.

4.2 Quadratic decay and intermediate lengths

Once both defects are small, one common bound suffices. Put

$$G(u) = \frac{1}{2} \left((1-u)^{-1} + (1-u)^{-2} + (1-u)^{-3} \right)^2. \quad (4.8)$$

This function is increasing on $[0, 1)$, and

$$f(1 - (1-u)^3) = G(u)u^2. \quad (4.9)$$

Proposition 4.3 (Uniform gap from two initial purities). *For every even $L \geq 4$, let H_L be a non-scalar finite-dimensional self-adjoint Hamiltonian. Suppose that $a \in \mathbb{R}$ and, for each $b > 0$, a real finite or countable list $(\lambda_i(b))$ satisfy*

$$\sum_i |\lambda_i(b)|^4 < \infty, \quad Z_L(b) := \text{Tr } e^{-b(H_L + aLI)} = \sum_i |\lambda_i(b)|^L \quad (L \geq 4 \text{ even}).$$

Define S, T by (4.1) and G by (4.8). Let $n_0 \geq 4$ be even, $b_0 > 0$, and let u_0, C satisfy

$$0 < u_0 < \frac{1}{6}, \quad G(u_0) \leq C, \quad r := Cu_0 < 1, \quad C(1 - 3u_0) > 3. \quad (4.10)$$

Suppose

$$S(n_0, b_0) \geq 1 - u_0, \quad T(2n_0, b_0) \geq 1 - u_0. \quad (4.11)$$

For $j \geq 0$, set

$$n_j = 2^j n_0, \quad b_j = 2^j b_0, \quad u_j = C^{-1} r^{2^j}. \quad (4.12)$$

Then

$$S(n_j, b_j) \geq 1 - u_j, \quad T(2n_j, b_j) \geq 1 - u_j. \quad (4.13)$$

For every even $L \geq n_0$, the physical ground state is unique, and the gap above it satisfies

$$\gamma_L > \frac{\log(1/r)}{b_0}. \quad (4.14)$$

Proof. Doubling. The definition gives $u_{j+1} = Cu_j^2$ and $0 < u_{j+1} < u_j \leq u_0$. Suppose both defects at (n_j, b_j) are at most $u = u_j$. The first bound in Lemma 4.2 is at most

$$f(1 - (1-u)^3) = G(u)u^2 \leq Cu^2 = u_{j+1} \leq u.$$

Use $p_+ = u_{j+1}$ in the second update. Its input is at most $1 - (1-u)^3$, so its output is also at most $Cu^2 = u_{j+1}$. Induction proves (4.13).

Intermediate even lengths. For any nonnegative list (a_i) with finite positive p th-power sum and $q \geq p > 0$, normalization gives

$$\sum_i a_i^q \leq \left(\sum_i a_i^p \right)^{q/p}.$$

Indeed the normalized p th powers are probabilities, whose (q/p) th powers sum to at most one. Apply this to the absolute spatial eigenvalues at each of the two temperatures. Whenever $n \leq L \leq 2n$ and both lengths are even,

$$Z_L(b) \leq Z_n(b)^{L/n}, \quad Z_L(2b) \geq Z_{2n}(2b)^{L/(2n)}.$$

Combining them yields the interpolation inequality

$$T(L, b) \geq \left[T(2n, b) S(n, b)^2 \right]^{L/(2n)}. \quad (4.15)$$

The bracket lies in $(0, 1]$ and the exponent lies in $[1/2, 1]$. For $n = n_j$ and $b = b_j$, we therefore have

$$T(L, b_j) \geq (1 - u_j)^3 \geq 1 - 3u_j > \frac{1}{2} \quad (n_j \leq L \leq 2n_j, \text{ } L \text{ even}). \quad (4.16)$$

The physical spectrum. The largest weight in (4.4) is at least its purity, since $\sum_k w_k^2 \leq (\max_k w_k) \sum_k w_k$. It is consequently greater than $1/2$. These weights count all physical eigenvectors; two ground-state eigenvectors would have two equal maximal weights, which is impossible. Thus the ground state is unique.

Because H_L is non-scalar and self-adjoint in finite dimension, a next distinct energy exists. Let w_0 be the ground-state weight and let w_1 be the weight of an eigenvector at that next energy. Then

$$\begin{aligned} e^{-b_j \gamma_L} &= \frac{w_1}{w_0} \leq \frac{1 - w_0}{w_0} \leq \frac{3u_j}{1 - 3u_j} \\ &\leq \frac{3}{C(1 - 3u_0)} r^{2^j} < r^{2^j}. \end{aligned} \quad (4.17)$$

Here the shift aL cancels, and the final strict inequality follows from (4.10). Taking logarithms and using $b_j = 2^j b_0$ proves (4.14) throughout the interval $n_j \leq L \leq 2n_j$. These dyadic intervals cover every even $L \geq n_0$. $\square$

Figure 1 summarizes the two operations in the proof: doubling advances the pair of purity estimates, and interpolation fills the interval of physical lengths controlled at each temperature.

4.3 The two parameter choices

We will establish the initial inequalities (4.11) for the following two rows:

n_0	b_0	u_0	C	$r = Cu_0$
60	105/4	1/8	79/10	79/80
2304	784	1/100	5	1/20

(4.18)

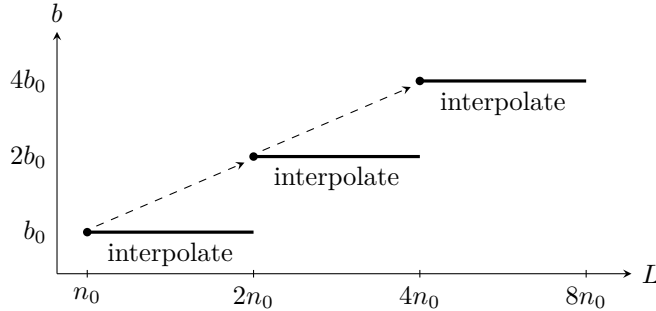


Figure 1: The bootstrap, shown schematically at three dyadic scales. Dashed arrows double the starting length and inverse temperature. At each scale, interpolation controls the physical purity for every even length on the solid interval. The intervals cover all larger even lengths.

The scalar hypotheses of Proposition 4.3 can already be checked. For the first row,

$$G(1/8) = \frac{913952}{117649} < \frac{79}{10}, \quad \frac{3}{C(1-3u_0)} = \frac{48}{79} < 1.$$

For the second, $1 - (1-u)^3 \leq 3u$ gives, when $0 < u \leq 1/100$,

$$G(u) \leq \frac{9}{2(1-3u)^2} \leq \frac{45000}{9409} < 5, \quad \frac{3}{C(1-3u_0)} = \frac{60}{97} < 1.$$

Thus the two rows give respectively the bounds $(4/105) \log(80/79)$ and $(\log 20)/784$ in (4.14), once their initial purities are proved. The second row will follow from six applications of Lemma 4.2 to bounds based at $(36, 49/4)$. The remaining task is therefore finite: establish the two sets of initial inequalities without losing the margins needed in these updates.

5 Two finite initializations

The doubling argument requires a spatial purity at length n and a physical purity at length $2n$, at the same inverse temperature. We establish two such pairs. The first begins at length 60; the second begins at length 2304 and gives the stronger eventual gap. In both cases the inputs are a few short-chain partition functions and one periodic trial vector. This section explains how those finite inputs imply the required purities.

5.1 Thermal data and elementary moment bounds

We use the sectors from Lemma 3.2: $J_n = I_n + 2O_n$ is the D_2 -invariant moment, I_n is the moment fixed also by the cyclic rotation, and N_n is the moment in one nontrivial D_2 sector. Each O_n occurs twice and each N_n occurs three times in the full trace. In particular,

$$\begin{aligned} J_n &= \frac{Z_n + 3Z_n(P)}{4}, & N_n &= \frac{Z_n - Z_n(P)}{4}, \\ I_n &= \frac{Z_n + 3Z_n(P) + 8Z_n(C)}{12}, & O_n &= \frac{Z_n + 3Z_n(P) - 4Z_n(C)}{12}. \end{aligned} \tag{5.1}$$

The inverse temperature is common to every term of each identity.

Lemma 5.1 (Finite thermal inputs). *For $a = 700741/500000$, the following tables give centers for the shifted partition functions $Z_n(b, g)$ in units 10^{-6} . The absolute errors are less than 10^{-5} at $b = 21/2$ and less than $30 \cdot 10^{-6}$ at $b = 49/4$.*

				$b = 49/4$		
				n	$g = 1$	$g = P$
				4	124885379	65757
				5	2925	2960742
				6	12870614	286599
				7	54069	1920414
				8	4639283	510140
				9	195455	1514846
				10	2654823	675419
				11	377629	1314544
				12	1900547	787405

				$b = 21/2$		
n	$g = 1$	$g = P$	$g = C$			
6	8945303	346734	939987			
8	3741281	567167	984108			
10	2327547	721090	1002613			

Appendices A and B prove the respective tables by rational approximations to the physical matrix exponentials, with explicit error bounds. Lemma C.2 supplies the other finite input:

$$E_0(72) < -72a, \quad E_0(120) < -120a. \quad (5.2)$$

Consequently $Z_{72}(b) > 1$ and $Z_{120}(b) > 1$ for every $b > 0$: in each case one Boltzmann factor already exceeds one.

Here are two elementary ways to extract spectral information from the tables.

Lemma 5.2 (Capped masses). *Let (t_i) be a finite or countable family with $0 \leq t_i \leq h$ and $\sum_i t_i \leq m$, where $h > 0$ and $m \geq 0$. For $r \geq 1$, put*

$$\mathcal{C}(m, h, r) = qh^r + (m - qh)^r, \quad q = \lfloor m/h \rfloor. \quad (5.3)$$

Then $\sum_i t_i^r \leq \mathcal{C}(m, h, r)$. Without the individual cap, $\sum_i t_i^r \leq m^r$.

Proof. For a finite family at fixed total mass, transfer mass from the smaller of two coordinates in $(0, h)$ to the larger until one reaches 0 or h . Convexity shows that the sum of r th powers does not decrease. Repetition leaves at most one coordinate in $(0, h)$, giving (5.3) at the actual total mass. That expression is continuous and nondecreasing in the total mass, so it is bounded by its value at m . Apply this argument to finite subfamilies and take their increasing union for a countable family. The uncapped assertion follows directly from $t_i^r \leq t_i(\sum_j t_j)^{r-1}$ when the total mass is positive; the zero case is immediate. $\square$

Lemma 5.3 (Polynomial filter). *Let $\epsilon \geq 0$, $v > 0$, and let (λ_i) be a real eigenvalue list with moments $M_j = \sum_i \lambda_i^j$, and suppose $|M_j - c_j| \leq \epsilon$ for the indices occurring in a polynomial $F(x) = \sum_j f_j x^j$. Assume these moments converge absolutely and $F \geq 0$ on $\mathbb{R}$. If*

$$B = \sum_j f_j c_j + \epsilon \sum_j |f_j| \quad \text{and} \quad F(x) > B \text{ for } |x| \geq v,$$

then every $|\lambda_i| < v$.

Proof. Absolute convergence gives $\sum_i F(\lambda_i) \leq B$. Since every summand is nonnegative, one eigenvalue with $|\lambda_i| \geq v$ would contradict this bound. $\square$

All moments used below converge absolutely by Proposition 3.1. The coefficient sums in (5.1) are at most one in absolute value, so the thermal error bound also bounds each sector-moment error. This observation includes the odd moments used in the second initialization; their signs are retained in the polynomial filter.

5.2 The initialization at length sixty

Proposition 5.4. *At inverse temperature $b = 105/4$,*

$$S(60, b) > \frac{7}{8}, \quad T(120, b) > \frac{7}{8}.$$

Proof. We first bound sector moments and then increase the inverse temperature. The trial energy forces a dominant spatial weight, giving the spatial purity; its physical Boltzmann factor will give the physical purity separately.

Begin at the lower inverse temperature $b_0 = 21/2$. From the $b_0 = 21/2$ table in Lemma 5.1 and (5.1), upper bounds for I_{10}, O_{10}, N_{10} are, respectively,

$$m_I = \frac{1042655}{10^6}, \quad m_O = \frac{40041}{10^6}, \quad m_N = \frac{401625}{10^6}.$$

For clarity, their margins over the sector centers plus 10^{-5} are $19/(12 \cdot 10^6)$, $7/(12 \cdot 10^6)$ and $3/(4 \cdot 10^6)$.

Apply Lemma 5.3 to

$$F_I(x) = x^6(-2 + 10x^2)^2, \quad F_N(x) = x^6(-25 + 100x^2)^2.$$

For a filter $x^6(c_0 + c_1x^2)^2$, its summed upper bound is the centered combination of moments 6, 8, 10 with coefficients $c_0^2, 2c_0c_1, c_1^2$, plus $10^{-5}(|c_0| + |c_1|)^2$. Direct rational evaluation gives

$$|\lambda| < U := \frac{100245}{100000} \quad \text{in } I, \quad |\lambda| < V := \frac{88643}{100000} \quad \text{in } N.$$

Indeed the filter value at the indicated endpoint exceeds its summed upper bound by more than 0.0301 for I and 0.3463 for N . Each filter is strictly increasing with $|x|$ beyond its endpoint: there $c_1 > 0$ and $c_0 + c_1x^2 > 0$.

The rational inequalities $U^{10} < m_I < 2U^{10}$ and $V^{10} < m_N < 2V^{10}$ allow the two-piece form of Lemma 5.2. For $k = 30, 120$ define

$$A_k = U^k + (m_I - U^{10})^{k/10}, \quad B_k = m_O^{k/10}, \quad C_k = V^k + (m_N - V^{10})^{k/10}.$$

They bound I_k, O_k, N_k , respectively. Because these are even moments, all are nonnegative, and hence

$$Z_k(b_0) \leq A_k + 2B_k + 3C_k, \quad Z_k(b_0, P), \quad Z_k(b_0, C) \leq A_k + 2B_k. \quad (5.4)$$

Put $m = A_{30} + 2B_{30} + 3C_{30}$, $p = A_{30} + 2B_{30}$, and $q = A_{120} + 2B_{120} + 3C_{120}$. With

$$D = \frac{1450}{1000}, \quad E = \frac{1202}{1000}, \quad \tau = \frac{1225}{1000}, \quad \tau_0 = \frac{1264}{1000}, \quad R_q = \frac{684}{10000},$$

the remaining scalar estimates are

$$\begin{aligned} m^5 < D^2, \quad p^5 < E^2, \quad q \geq 1, \quad (q-1)^5 < R_q^2, \\ \frac{D+11E}{12} \leq \tau, \quad \frac{D+3E}{4} \leq \tau_0. \end{aligned} \quad (5.5)$$

For example, the positive margins in the first, second and fourth strict inequalities exceed 0.001068, 0.000433 and 0.0000472, respectively. Every number in these comparisons is rational and is specified above.

We now increase the inverse temperature from b_0 to $b = (5/2)b_0$. For positive Boltzmann factors z_i , $\sum_i z_i^{5/2} \leq (\sum_i z_i)^{5/2}$. Thus (5.4)–(5.5) imply

$$Z_{30}(b) \leq D, \quad Z_{30}(b, P), Z_{30}(b, C) \leq E.$$

By (5.1), the I and J portions of the thirtieth spatial moment are at most τ and τ_0 .

Let $w_i = |\lambda_i(X_b)|^{30}$. Then $\sum_i w_i \leq D$, whereas (5.2) gives $\sum_i w_i^4 = Z_{120}(b) > 1$. Since $D(0.88)^3 < 1$, some weight $w_0 > 0.88$ exists. It is unique because $2(0.88) > D$. It belongs to I : a

weight in O would occur at least twice, and one in N at least three times. These statements count all multiplicities, including equal moduli of opposite-sign spatial eigenvalues.

To sharpen the lower bound on w_0 , suppose $0 \leq t \leq 0.998$ and $w_0 \geq t$. Denote the remaining I mass by x , the full O -pair mass by y , and the full N -triple mass by z . Their squared weights sum to at most $x^2 + y^2/2 + z^2/3$, and

$$0 \leq x \leq \tau - t, \quad 0 \leq y, \quad x + y \leq \tau_0 - t, \quad z \leq D - t - x - y.$$

The divisors 2 and 3 use the identical weight lists in the repeated sectors. Since $D > \tau_0$, the last upper bound is nonnegative on the entire displayed polygon. The convex quadratic $x^2 + y^2/2 + (D - t - x - y)^2/3$ has its maximum at one of its four vertices. Write

$$R(t) = \max_{(x,y) \in \mathcal{V}_t} \left(x^2 + \frac{y^2}{2} + \frac{(D - t - x - y)^2}{3} \right), \quad (5.6)$$

where

$$\mathcal{V}_t = \{(0,0), (\tau - t, 0), (\tau - t, \tau_0 - \tau), (0, \tau_0 - t)\}.$$

Thus the remaining squared weights sum to at most $R(t)$, and their fourth powers sum to at most $R(t)^2$. Exact evaluation gives

t	0.88	0.99	0.998
$R(t)$	0.1359	0.0721	0.068404

and

$$1 - 0.99^4 - R(0.88)^2 = 0.02093518, \quad 1 - 0.998^4 - R(0.99)^2 = 0.002777621984.$$

If $w_0 \leq 0.99$, the first equality would contradict $\sum_i w_i^4 > 1$; the second then excludes $w_0 \leq 0.998$. Keeping the dominant contribution in the fourth moment yields

$$S(60, b) = \frac{\sum_i w_i^4}{(\sum_i w_i^2)^2} \geq \left(1 + \frac{R(0.998)}{0.998^2} \right)^{-2} > 0.8756 > \frac{7}{8}.$$

For the physical purity at length 120, return briefly to temperature b_0 . Its largest Boltzmann factor z_0 exceeds one by the trial bound, while the sum of all factors is at most q . After raising the factors to the power $5/2$, the ratio of the remaining mass to $z_0^{5/2}$ is at most $(q - 1)^{5/2} < R_q$. Therefore

$$T(120, b) \geq (1 + R_q)^{-2} > 0.8760 > \frac{7}{8}.$$

This last step concerns physical Boltzmann factors, separately from the spatial weights used to bound S . □

5.3 The initialization giving the stronger gap

Proposition 5.5. *At inverse temperature $b = 784$,*

$$1 - S(2304, b) \leq \frac{84513}{10^7} < \frac{1}{100}, \quad 1 - T(4608, b) \leq \frac{27493}{5 \cdot 10^6} < \frac{1}{100}.$$

Proof. We isolate one large eigenvalue in J and bound the remaining moments. This gives a pair of initial purity defects at length 36; six rounded updates then reach length 2304.

Begin at $b_0 = 49/4$. The $b_0 = 49/4$ table gives centers for J_n, N_n through (5.1), with error at most $\epsilon = 30 \cdot 10^{-6}$. Use the two filters

$$F_J(x) = x^4(7 + 17x - 280x^2 - 185x^3 + 1000x^4)^2, \quad F_N(x) = x^4(12 - 394x^2 + 1000x^4)^2.$$

Their degree is 12, so the table supplies every required moment. The summed upper bounds in Lemma 5.3 are exactly

$$B_J = 318604.54848175, \quad B_N = 48169.880699.$$

Set $v_J = 100139/100000$ and $v_N = 872/1000$. Direct rational comparisons give

$$F_J(v_J) - B_J > 180, \quad F_J(-v_J) - B_J > 496888, \quad F_N(\pm v_N) - B_N > 654.$$

To check the whole exterior rays without further numerical tests, write $Q(x) = 7 + 17x - 280x^2 - 185x^3 + 1000x^4$. For $y \geq 0$,

$$\begin{aligned} Q(1+y) &= 559 + 2902y + 5165y^2 + 3815y^3 + 1000y^4, \\ Q(-1-y) &= 895 + 3978y + 6275y^2 + 4185y^3 + 1000y^4. \end{aligned}$$

Both are positive and increasing. The polynomial $12 - 394r^2 + 1000r^4$ is likewise positive and increasing for $r \geq v_N$: its value at v_N is positive and its derivative is $4r(1000r^2 - 197) > 0$. Thus the filters increase along the requisite rays, and every eigenvalue has modulus less than v_J in J and less than v_N in N .

The twelfth-moment upper bounds are

$$m_J = 1.0657205, \quad m_N = 0.2783155.$$

Put $\ell = 999/1000$ and, for $k = 36, 72$, define

$$B(k) = 3\mathcal{C}(m_N, v_N^{12}, k/12), \quad R_A(k) = B(k) + (m_J - \ell^{12})^{k/12}. \quad (5.7)$$

Here $m_J > \ell^{12}$. If all eigenvalues in J had modulus at most ℓ , Lemma 5.2 would imply

$$Z_{72}(b_0) \leq \mathcal{C}(m_J, \ell^{12}, 6) + B(72) < 1,$$

where the strict margin to one exceeds 0.0693. The trial inequality contradicts this. Choose an eigenvalue λ_0 in J with $|\lambda_0| > \ell$. Removing it leaves a twelfth-moment mass at most $m_J - \ell^{12}$ in J . The uncapped bound of Lemma 5.2 and the three N sectors show that $R_A(k)$ bounds the remaining full k th moment.

It follows that

$$\begin{aligned} 1 - S(36, b_0) &\leq p_0 := 1 - \left(1 + \frac{R_A(36)}{\ell^{36}}\right)^{-2}, \\ 1 - T(72, b_0) &\leq q_0 := 1 - \left(v_J^{72} + R_A(72)\right)^{-2}. \end{aligned} \quad (5.8)$$

For the first estimate retain λ_0^{72} in the numerator; for the second use $Z_{72}(2b_0) > 1$ and $Z_{72}(b_0) \leq v_J^{72} + R_A(72)$. Numerically $p_0 < 0.047916$ and $q_0 < 0.181521$, but the following updates use their exact rational definitions in (5.7)–(5.8).

Apply Lemma 4.2 six times, rounding each new defect upward to a multiple of 10^{-7} and using the rounded spatial defect in the physical update. For completeness, with $f(u) = u^2/[2(1-u)^2]$ and $\lceil x \rceil_7 = 10^{-7} \lceil 10^7 x \rceil$, the scalar updates are

$$p_{j+1} = \left\lceil f(1 - (1 - p_j)^2(1 - q_j)) \right\rceil_7, \quad q_{j+1} = \left\lceil f(1 - (1 - q_j)^2(1 - p_{j+1})) \right\rceil_7.$$

Exact rational evaluation gives

j	p_j	q_j
1	151249/2500000	433453/2500000
2	343303/5000000	1632381/10000000
3	44601/625000	361773/2500000
4	632939/10000000	1055161/10000000
5	375793/10000000	445941/10000000
6	84513/10000000	27493/5000000

Every update is in $[0, 1)$, where f is increasing, so the upward rounding preserves valid bounds. Both the length and inverse temperature have multiplied by 2^6 : $36 \cdot 64 = 2304$ and $(49/4) \cdot 64 = 784$. The last row proves the proposition. $\square$

5.4 A periodic interface for boundary estimates

The boundary-field companion [19] uses the intermediate spatial information in the preceding proof, rather than only the final periodic gap. We state that information separately so that its hypotheses and finite inputs can be invoked together.

Corollary 5.6 (Periodic inputs for boundary estimates). *Let X_b be the spatial transfer operator of Proposition 3.1, with the shift $a = 700741/500000$, and put*

$$\begin{aligned} b_0 &= 49/4, & b_1 &= 49/2, \\ \ell &= 999/1000, & U &= 100139/100000, & V &= 872/1000, \\ m_J &= 1.0657205, & m_N &= .2783155. \end{aligned}$$

At $b = b_0$, the full eigenvalue list, counted with multiplicity, is the D_2 -invariant list J together with three identical lists N . Every eigenvalue in J has modulus less than U , and the sum of the twelfth powers of all eigenvalues in J is at most m_J . Every eigenvalue in each N list has modulus less than V , and that list's twelfth powers sum to at most m_N . There is an eigenvalue in J whose modulus exceeds ℓ . Moreover,

$$1 - S(72, b_1) \leq \frac{151249}{2500000}.$$

For $n = 72$ and $n = 120$, one has $E_0(n) + na < 0$; hence a physical ground Boltzmann factor in $Z_n(b)$ exceeds one for every $b > 0$.

Proof. The decomposition is Lemma 3.2, with $J = I + 2O$. The filters, twelfth-moment bounds and dominant eigenvalue are proved at the start of the proof of Proposition 5.5. The first row of its rounded update table gives the displayed bound at $(72, 49/2)$. The variational assertion is Lemma C.2. $\square$

6 Proof of the gap theorem

We now assemble the two initializations through the same purity theorem. The thermal and trial inputs used in Section 5 are proved in Appendices A–C. Those finite proofs are part of the argument.

Proof of Theorem 1.1. Proposition 3.1 identifies the physical partition functions with moments of the self-adjoint spatial transfer operator. They therefore satisfy the hypotheses used to define both purities in Section 4. The Hamiltonians are self-adjoint and non-scalar, as noted after (1.2).

Proposition 5.4 supplies

$$S(60, 105/4) > 7/8, \quad T(120, 105/4) > 7/8.$$

Apply Proposition 4.3 with

$$(n_0, b_0, u_0, C) = (60, 105/4, 1/8, 79/10).$$

Its parameter inequalities are verified in Section 4, and $Cu_0 = 79/80$. It follows that the ground state is unique and $\gamma_L > \frac{4}{105} \log(80/79)$ for every even $L \geq 60$.

Proposition 5.5 supplies both purity defects at most $1/100$ at the pair based at $(n_0, b_0) = (2304, 784)$. Apply the same theorem with $u_0 = 1/100$ and $C = 5$. Since $Cu_0 = 1/20$, it gives $\gamma_L > \log(20)/784$ for every even $L \geq 2304$. Taking the limit inferior over even lengths proves the stated bound on Δ_1 . Finally, Lemma 2.1 transfers the spectral conclusions to the other spin-one matrix realization by simultaneous similarity and its tensor powers. $\square$

6.1 An infinite-volume consequence

The finite-volume estimate also controls local excitations in a thermodynamic limit. We use the standard limiting argument recalled in [22, arXiv v4, End Matter, Definition 11 and Theorem 12]. A local operator on the spin chain $\mathbb{Z}$ acts nontrivially on only finitely many sites. A state ω is a linear functional on these operators with $\omega(I) = 1$ and $\omega(A^*A) \geq 0$ for every local operator A . Define the local commutator

$$[H, A] = \sum_{j \in \mathbb{Z}} [\mathbf{S}_j \cdot \mathbf{S}_{j+1}, A].$$

Only finitely many summands are nonzero.

Corollary 6.1. *For the self-adjoint Hamiltonians (1.1), the normalized ground states of the even periodic rings have a subsequence, as $L \rightarrow \infty$, converging on every local operator to a state ω . Every such limit satisfies*

$$\omega(A^*[H, A]) \geq \frac{\log 20}{784} (\omega(A^*A) - |\omega(A)|^2) \quad (6.1)$$

for every local operator A .

Proof. Place the ring of even length L on $\{-L/2, \dots, L/2 - 1\}$, with its closing bond at the endpoints. Compactness of the density matrices on each finite interval and a diagonal subsequence give convergence on all local operators. The compatible limits define a positive normalized functional ω . For $L \geq 2304$, Theorem 1.1 gives a unique ground state, with normalized vector ψ_L , and a gap greater than $\delta = \log(20)/784$. Writing $\omega_L(A) = \langle \psi_L, A\psi_L \rangle$, the spectral inequality on the orthogonal complement of ψ_L gives

$$\omega_L(A^*[H_L, A]) \geq \delta (\omega_L(A^*A) - |\omega_L(A)|^2).$$

For fixed local A and sufficiently large L , the closing bond commutes with A and $[H_L, A] = [H, A]$. Passing to the local limit proves the claim. $\square$

In particular, ω is a ground state: no local perturbation lowers its energy. For $\omega(A) = 0$, inequality (6.1) is the locally unique gapped property of [22, arXiv v4, Definition 11], with gap at least $\log(20)/784$. The argument does not identify the limits of different subsequences or establish uniqueness among all infinite-volume ground states.

A Thermal calculation at inverse temperature $21/2$

We prove the $b = 21/2$ table in Lemma 5.1. The calculation uses the actual physical Hamiltonians at lengths $n = 6, 8, 10$, with twists $1, P, C$. All arithmetic below is integral or rational; in the cyclic case complex numbers are represented in $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3}$.

A.1 The finite matrices and their columns

Fix $b = 21/2$ and one of these lengths, and write $H = H_n(g)$. After the global rotations in Section 2, the three twists have angles $\theta = 0, \pi, 2\pi/3$ about the weight-basis axis. Set

$$W = I - \frac{H_n(g) + (3n/2)I}{16}, \quad B = 16W. \quad (\text{A.1})$$

Lemma 2.2 gives $\|W\| \leq 1$ and $\|B\| \leq 16$. In a fixed total-weight sector, the rows and columns are words $x \in \{-1, 0, 1\}^n$ with fixed $\sum_j x_j$. The diagonal of B is

$$B_{x,x} = 16 - \frac{3n}{2} - \sum_{j=0}^{n-1} x_{j-1}x_j.$$

For each j , put $k = j + 1 \pmod{n}$. If $y_j = x_j - \delta$, $y_k = x_k + \delta$, $\delta \in \{-1, 1\}$, and all letters remain in $\{-1, 0, 1\}$, the corresponding entry is -1 , except at $j = n - 1, k = 0$, where it is $-e^{i\theta\delta}$. This is exactly the physical seam phase in (2.5). For the cyclic twist its action on a coordinate $u + v\omega$ uses

$$-\omega(u + v\omega) = v + (v - u)\omega, \quad -\omega^{-1}(u + v\omega) = (u - v) + u\omega. \quad (\text{A.2})$$

Thus matrix-vector multiplication requires only integer additions and multiplications in both coordinates.

Only some columns need to be computed. To justify the reduction, let $s(x) = (x_{n-1}, x_0, \dots, x_{n-2})$. The seam formula gives

$$H_{s(x), s(y)} = e^{i\theta(y_{n-1} - x_{n-1})} H_{x, y}.$$

Consequently the monomial unitary $|x\rangle \mapsto e^{-i\theta x_{n-1}} |s(x)\rangle$ commutes with H . Reversal of the word takes H to its complex conjugate, as does negation of every letter. Composing either permutation with complex conjugation gives an antiunitary symmetry. Each of these symmetries preserves the norm of the corresponding column of any real exponential of H .

In every sector of nonnegative total weight, select the lexicographically least word in each orbit under rotations and reversal. Give it the orbit size as its multiplicity, doubled for strictly positive total weight to include the opposite sector. The multiplicities then sum to 3^n and reproduce the full Frobenius norm of the exact exponential. No symmetry of the rounded columns is required: we compare each computed representative with its own exact column and repeat that comparison with the same multiplicity.

A.2 A rational polynomial and its error

Write

$$E = e^{84(W-I)} = e^{-(b/2)(H_n(g) + (3n/2)I)}, \quad x_n = (3/2 - a)bn.$$

The desired trace is

$$Z_n(b, g) = e^{x_n} \|E\|_F^2, \quad (\text{A.3})$$

where $\|\cdot\|_F$ is the Frobenius norm. For $0 \leq j \leq J = 256$, define the rational probabilities

$$p_j = \frac{84^j/j!}{\sum_{l=0}^{256} 84^l/l!}, \quad P(W) = \sum_{j=0}^{256} p_j W^j. \quad (\text{A.4})$$

These are Poisson probabilities of mean 84, conditioned on values at most 256. For a Poisson variable V with that mean,

$$\Pr(V > 256) \leq \frac{\mathbb{E}(2^V)}{2^{257}} = \frac{e^{84}}{2^{257}} < \frac{27^{28}}{2^{257}} < 2^{-117}.$$

The conditioning estimate is the coefficient argument of Fox–Glynn [10, Proposition 1], applied here to powers of a contraction. Since $\|W^j\| \leq 1$, conditioning changes the operator average by at most twice this probability. Hence

$$\|P(W) - E\| < 2^{-116}. \quad (\text{A.5})$$

Put $Q = 2^{56}$ and $q_j = \lfloor Q p_j \rfloor$. For a representative column r , start at zero and apply, for $j = 256, \dots, 0$,

$$z \leftarrow \text{fl}(Bz/16) + q_j e_r. \quad (\text{A.6})$$

For real coordinates, fl is the ordinary floor; for $u + v\omega$ it floors u and v separately. The ideal suffix at step j is $Q \sum_{l=j}^{256} p_l W^{l-j} e_r$, whose norm is at most Q . In a sector of dimension d_0 , the Euclidean norm of the error from a coordinatewise floor is less than $2\sqrt{d_0}$, and the coefficient floor adds at most one. Contraction of W therefore bounds the error at every stage by

$$257(2\sqrt{3^n} + 1) \leq 125159. \quad (\text{A.7})$$

The zero coefficients above q_{172} may be omitted: the computed vector is still zero there, and (A.7) continues to count all 257 possible rounding steps.

The norm of the repeated computed-column list, divided by Q and multiplied by $e^{x_n/2}$, differs from $\sqrt{Z_n(b, g)}$ by at most

$$e^{x_n/2} \sqrt{3^n} \left(\frac{257(2\sqrt{3^n} + 1)}{Q} + 2^{-116} \right) < 4 \cdot 10^{-7}. \quad (\text{A.8})$$

Indeed $\sqrt{3^n} \leq 243$, $x_n/2 < 6$, and $e^{x_n/2} < 3^6$. The resulting rational upper bound $3^6 243(125159/2^{56} + 2^{-116})$ is less than $4 \cdot 10^{-7}$. This comparison includes both polynomial truncation and integer rounding.

A.3 Exact evaluation by packed integers

For efficiency, the calculation performs (A.6) on batches of at most 192 columns. A list of signed integer digits $y_0, \dots, y_{w-1}$ is stored as

$$\text{pack}(y) = \sum_{t=0}^{w-1} y_t 2^{64t}.$$

Integer linear arithmetic preserves this identity before the floor, even if intermediate carries occur. Arbitrary-precision integers are used throughout.

Here is why the floor can also be performed on the packed integer. For either coefficient of $u + v\omega$,

$$|u|, |v| \leq \frac{2}{\sqrt{3}} |u + v\omega| \leq 2|u + v\omega|.$$

By (A.7), every computed column has norm at most $Q + 125159$. Thus every coordinate of Bz , in either integer component, has magnitude at most

$$32(Q + 125159) = 2305843009217699040 < 2^{63}. \quad (\text{A.9})$$

Let $s_w = \sum_{t=0}^{w-1} 2^{64t}$ and define

$$\text{bias} = 2^{63}s_w, \quad \text{mask} = (2^{60} - 1)s_w, \quad \text{bias16} = 2^{59}s_w.$$

For $Y = \text{pack}(y)$ satisfying (A.9), the operation

$$(((Y + \text{bias}) \gg 4) \& \text{mask}) - \text{bias16}$$

equals $\text{pack}(\lfloor y_t/16 \rfloor)_{t=0}^{w-1}$. To see this, adding the bias makes every base- 2^{64} digit lie in $[0, 2^{64})$. The global right shift moves four low bits from each higher lane into the four high bits of the preceding lane. The mask deletes precisely those crossing bits, retaining $\lfloor (y_t + 2^{63})/16 \rfloor$ in lane t . Subtracting 2^{59} then gives $\lfloor y_t/16 \rfloor$, also for negative y_t . This proves by induction that the packed recurrence is exactly (A.6).

The final coordinates satisfy the smaller bound $2(Q + 125159) < 2^{63}$, so biasing and unpacking recovers every integer coordinate exactly. If a final coordinate is $u + v\omega$, its squared modulus is $u^2 - uv + v^2$. Summing these integers with the representative multiplicities gives an integer $t_{n,g}$, the squared norm of the full repeated-column list in Q -units.

A.4 The exact totals and trace enclosures

Integer evaluation of the recurrence gives the following totals. The accompanying computation implements the matrix entries, representatives and packed operations specified above.

n	g	$t_{n,g}$
6	1	93637037510293158533967385845216
6	P	3629519998654831111446492683663
6	C	9839536903588146388838689926305
8	1	4947327904487991279560670127283
8	P	749999795795292705866802820621
8	C	1301346729026243689610427021457
10	1	388818111664728958208623998490
10	P	120458568770462253326886628639
10	C	167487042402286114495695779656

To enclose the scalar factor, put

$$L_n = \sum_{j=0}^{100} \frac{x_n^j}{j!}, \quad U_n = L_n + 2 \frac{x_n^{101}}{101!}.$$

Since $0 < x_n < 12$ and successive terms after the first omitted term have ratio at most $x_n/102 < 1/2$, we have $L_n < e^{x_n} < U_n$. For each center $c_{n,g}$ in the $b = 21/2$ table of Lemma 5.1, exact rational comparison gives

$$c_{n,g} - 2 \cdot 10^{-6} < \frac{L_n t_{n,g}}{Q^2} < \frac{U_n t_{n,g}}{Q^2} < c_{n,g} + 2 \cdot 10^{-6}. \quad (\text{A.10})$$

Here the centers are interpreted as real numbers, including their 10^{-6} scale. These inequalities in particular put every computed scaled norm below 4. By (A.8), replacing its square by the true trace incurs error less than $(8 + 4 \cdot 10^{-7})4 \cdot 10^{-7}$. Together with (A.10), this yields

$$|Z_n(21/2, g) - c_{n,g}| < 2 \cdot 10^{-6} + (8 + 4 \cdot 10^{-7})4 \cdot 10^{-7} < 10^{-5}.$$

This proves the $b = 21/2$ table in Lemma 5.1.

B Thermal calculation at inverse temperature 49/4

We prove the $b = 49/4$ table in Lemma 5.1. The computation uses integer approximations to columns of a physical matrix exponential. We specify the integer algorithm, bound its error, and give its exact outputs. Throughout this section,

$$b = \frac{49}{4}, \quad J = 280, \quad Q = 2^{55}, \quad 4 \leq n \leq 12.$$

The twists are the identity and a rotation by π about the z axis. The latter has the same partition function as P , by the global rotation described in Section 2.

B.1 The physical matrix and its rational approximation

In the weight basis, write a word as $w = (w_0, \dots, w_{n-1})$, with $w_i \in \{-1, 0, 1\}$. The total weight $\sum_i w_i$ is conserved. In each weight block let H be the restriction of $H_n(g)$, and set

$$R = I - \frac{\frac{3}{2}nI + H}{16}, \quad D = 32R. \tag{B.1}$$

By (2.5), D is an integer matrix with diagonal

$$D_{w,w} = 32 - 3n - 2 \sum_{i=0}^{n-1} w_{i-1} w_i.$$

An allowed opposite unit step on the letters at $i-1, i$ has entry -2 , except that the entry is $+2$ on the twisted seam $i=0$. These formulas specify every matrix entry; indices on words are modulo n . The unit step coefficient follows also directly from the two ladder factors $\sqrt{2}$ and the transverse prefactor $1/2$.

Lemma 2.2 gives

$$0 \leq \frac{3}{2}nI + H \leq \frac{5}{2}nI, \quad \text{spec}(R) \subseteq [1 - 5n/32, 1] \subseteq [-7/8, 1].$$

Thus R is a self-adjoint contraction. Put

$$F = \exp \left[-\frac{b}{2} \left(\frac{3}{2}nI + H \right) \right] = \exp[98(R - I)].$$

We use the same symbols for the direct sums over all weight blocks. On the full physical space,

$$Z_n(b, g) = e^{x_n} \|F\|_F^2, \quad x_n = bn \left(\frac{3}{2} - a \right), \tag{B.2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and the squared norms of the weight blocks are summed.

Define the rational probabilities

$$p_j = \frac{v_j}{\sum_{\ell=0}^J v_\ell}, \quad v_j = 98^j \frac{J!}{j!}, \quad 0 \leq j \leq J. \quad (\text{B.3})$$

They are Poisson weights of mean 98, conditioned to indices at most J . Since $\|R\| \leq 1$, discarding the Poisson tail and renormalizing changes the exponential by at most twice the tail probability, by the same coefficient estimate as in Fox–Glynn [10, Proposition 1]:

$$\left\| F - \sum_{j=0}^J p_j R^j \right\| \leq 2e^{-98} \sum_{j=281}^{\infty} \frac{98^j}{j!} \leq 2^{-280} e^{98} < 2^{-84} < Q^{-1}.$$

For the second inequality, use $2^j \geq 2^{281}$ in the tail and $e^{-98} \sum_{j \geq 0} 196^j / j! = e^{98}$; the next uses $e < 4$. The estimate includes the change of normalization.

B.2 Integer columns and their error

For a selected unit basis column e_r , start with $X = 0$ and perform, in the order $j = J, J-1, \dots, 0$,

$$X \leftarrow \left\lfloor \frac{DX}{32} \right\rfloor + c_j e_r, \quad c_j = \lfloor Q p_j \rfloor. \quad (\text{B.4})$$

The vector floor is coordinatewise. All quantities in this recursion are integers. The coefficients c_j vanish for $j > 191$, so those initial zero steps may be omitted without changing X .

To bound rounding, compare with the ideal suffix

$$Y_j = Q \sum_{\ell=j}^J p_\ell R^{\ell-j} e_r, \quad \|Y_j\|_2 \leq Q.$$

In a block of dimension d_0 , one vector floor introduces Euclidean error less than $\sqrt{d_0}$, and one coefficient floor error less than one. The contraction property therefore bounds the accumulated error by $281(\sqrt{d_0} + 1)$. Set

$$d_n = \lfloor \sqrt{3^n} \rfloor + 1, \quad E_n = 281(d_n + 1) + 1, \quad \mathcal{E}_n = d_n E_n. \quad (\text{B.5})$$

Including the preceding exponential-approximation error, every computed column satisfies

$$\|X - Q F e_r\|_2 \leq E_n. \quad (\text{B.6})$$

The bound charges all 281 ideal steps, including the omitted steps whose rational coefficients are nonzero but whose integer floors vanish.

The recursion can be evaluated exactly with signed 64-bit integers. Indeed every intermediate coordinate has absolute value at most

$$Q + 281(d_n + 1) < 2Q, \quad d_n \leq 730.$$

The absolute sum of any row of D is at most

$$|32 - 3n| + 2n + 4n \leq 76.$$

The first two terms bound the diagonal, and at most two hops per bond give the last term. Consequently every product and every partial sum in a sparse matrix multiplication has absolute value less than

$$76 \cdot 2Q = 5476377146882523136 < 2^{63} - 1.$$

Division by 32 and addition of $c_j \leq Q$ also fit; even their conservative absolute bound is

$$\frac{76 \cdot 2Q}{32} + Q = 207165582859042816 < 2^{63} - 1.$$

Starting with $X = 0$, these bounds first establish exactness of the next integer operation; the contraction estimate then establishes the next coordinate bound. Thus the induction does not assume the absence of overflow that it is proving. All final squares and sums below use arbitrary-precision integers.

B.3 Representative columns

Only one column from each of the following word orbits is needed. For positive total weight, use the orbit under cyclic shifts and reversal, and give its representative twice the orbit size. This includes the uncomputed negative-weight block. In the zero-weight block, include digit negation in the orbit itself and use its size without an extra factor two. Choose the lexicographically least word as representative. Set cardinalities handle periodic words and all other nontrivial stabilizers. The multiplicities ν_r satisfy

$$\sum_r \nu_r = 3^n.$$

Here are the exact symmetries underlying this reduction. Reversal $w \mapsto (w_{n-1}, \dots, w_0)$ fixes the seam as an unordered bond, and global negation preserves every Hamiltonian entry. For the twisted chain, a bare shift moves the seam. Left rotation by q moves it to $(k-1, k)$, where $k = -q \bmod n$. The diagonal unitary

$$G_k|w\rangle = (-1)^{\sum_{j=0}^{k-1} w_j} |w\rangle, \quad G_0 = I,$$

flips the hopping signs exactly at the two boundary bonds of this arc. It restores the original seam. The resulting signed shift commutes with H , so exact exponential columns at rotated words have equal norms. For the untwisted chain no sign correction is needed.

Let X_r be the output of (B.4) at representative r , and define the integer

$$W_{n,g} = \sum_r \nu_r \sum_w X_r(w)^2. \tag{B.7}$$

Rounding need not preserve the signed symmetries. To compare norms, form abstract direct sums repeating each exact representative column ν_r times, and do the same for each computed column. The first direct sum has squared norm $Q^2 \|F\|_F^2$, and the second has squared norm $W_{n,g}$. By (B.6),

$$\left| \sqrt{W_{n,g}} - Q \|F\|_F \right| \leq \sqrt{3^n} E_n < \mathcal{E}_n. \tag{B.8}$$

This proves the full trace estimate using representatives, without requiring their rounded columns to transform equivariantly.

B.4 Exact totals and rational enclosures

The integer algorithm (B.3), (B.4), and (B.7) gives the following totals. The second column uses no twist; the third uses the π twist.

n	$W_{n,I}$	$W_{n,P}$
4	1298092848946206529211955779616040	683496640248050713756440676071
5	9095379218684502088743241789	9205948219074287936685289758037
6	11971284263768914176431179720622	266573596119930013833919176645
7	15044150708248976451784593468	534330762492192297990602166370
8	386135473475360677201628357188	42459883501386986922766507327
9	4866417737994347551816846903	37716485750353424806876104310
10	1977296037421547282775892486	5030484777222468978572235564
11	841347687722481981930750335	2928767918985354021900689432
12	1266666171463993322695232916	524785463625387865619045258

We finish with a finite rational enclosure, specifying every bound used to turn these integers into the thermal table at $b = 49/4$. Let

$$r_{n,g} = \lfloor \sqrt{W_{n,g}} \rfloor, \quad s_n = \sum_{j=0}^{100} \frac{x_n^j}{j!}, \quad t_n = \frac{x_n^{100}}{100!}.$$

Since $0 \leq x_n < 15$, successive omitted terms have ratio at most $15/101$, and the tail is at most $15t_n/86 < t_n$. Hence $s_n \leq e^{x_n} \leq s_n + t_n$. Equations (B.2) and (B.8) give

$$L_{n,g} = \frac{s_n \max(0, r_{n,g} - \mathcal{E}_n)^2}{Q^2}, \quad U_{n,g} = \frac{(s_n + t_n)(r_{n,g} + 1 + \mathcal{E}_n)^2}{Q^2}, \quad (B.9)$$

$$L_{n,g} \leq Z_n(b, g) \leq U_{n,g}.$$

All quantities in these endpoints are rational or integer, with the integer square root determined by $r^2 \leq W < (r+1)^2$. Writing $C_{n,g}$ for the integer centers in the $b = 49/4$ table of Lemma 5.1, direct integer and rational evaluation yields

$$L_{n,g} > \frac{C_{n,g} - 30}{10^6}, \quad U_{n,g} < \frac{C_{n,g} + 30}{10^6} \quad (4 \leq n \leq 12, g = I, P).$$

Every one of these strict inequalities has margin greater than $13/10^6$. The displayed algorithm, totals, and enclosures give a reproducible finite verification of all eighteen enclosures, and complete the proof of the thermal table at $b = 49/4$.

C Two variational inputs

The finite initializations need a lower bound on the untwisted partition function at lengths 72 and 120. One fixed triple of integer matrices defines periodic matrix-product vectors at both lengths, with energy per bond below $-a = -700741/500000$. A block decomposition reduces the calculation to powers of integer matrices of order at most 162.

C.1 Norm and energy contractions

For $N \geq 4$ and three real D -by- D matrices A_s , indexed by $s = -1, 0, 1$, define a vector in the standard orthonormal spin-one weight basis by

$$\psi_N(s_0, \dots, s_{N-1}) = \text{Tr}(A_{s_0} \cdots A_{s_{N-1}}). \quad (C.1)$$

This is the periodic matrix-product representation of [20, Section 2.1, Equations (2)–(3)]. Write

$$B_{st} = A_s A_t, \quad T = \sum_{s=-1}^1 A_s \otimes A_s, \quad (\text{C.2})$$

$$O = \sum_{s,t=-1}^1 st B_{st} \otimes B_{st} + \sum_{\substack{s=-1,0 \\ t=0,1}} (B_{st} \otimes B_{s+1,t-1} + B_{s+1,t-1} \otimes B_{st}). \quad (\text{C.3})$$

All traces here are ordinary, unnormalized matrix traces.

Lemma C.1. *Let $N \geq 4$ and $D \geq 1$ be integers, and let $A_{-1}, A_0, A_1 \in M_D(\mathbb{R})$. Define ψ_N, T, O by Equations (C.1)–(C.3). For the periodic spin-one Heisenberg Hamiltonian H_N , set*

$$V_N = \text{Tr}(T^N), \quad U_N = \text{Tr}(T^{N-2}O).$$

Then

$$\|\psi_N\|^2 = V_N, \quad \langle \psi_N, H_N \psi_N \rangle = NU_N.$$

In particular, if $V_N > 0$ and $U_N < -aV_N$, then $E_0(N) < -Na$ and $Z_N(b) > 1$ for every $b > 0$.

Proof. Expand T^N and use $\text{Tr}(X \otimes Y) = \text{Tr}(X) \text{Tr}(Y)$. Since all amplitudes in (C.1) are real, the result is the sum of their squares. This proves the norm formula without assuming that the finite contraction matrix T is self-adjoint or positive.

In the weight basis, a bond has diagonal entry st on the pair (s, t) . Its off-diagonal entries are 1 between (s, t) and $(s+1, t-1)$ for $s = -1, 0$ and $t = 0, 1$: each follows from $h = S^z \otimes S^z + (S^+ \otimes S^- + S^- \otimes S^+)/2$ and the ladder entries $\sqrt{2}$. Inserting this bond into the two amplitude layers gives exactly O in (C.3). Summing the other physical indices therefore gives U_N for one bond. Cyclicity of the amplitude trace makes ψ_N invariant under cyclic site translation, so every bond, including the periodic seam, contributes equally.

For $V_N > 0$, the variational principle gives $E_0(N) \leq NU_N/V_N < -Na$. The shifted ground energy is negative, so its term in $\text{Tr} e^{-b(H_N + NaI)}$ exceeds 1 for every $b > 0$. $\square$

C.2 One integer matrix triple

Set $D = 26$. Index eight types by $\alpha = 1, \dots, 8$, put

$$(\ell_1, \dots, \ell_8) = (1, 1, 1, 1, 3, 3, 3, 5),$$

and order the virtual labels (α, d) first by α and then by $d = -\ell_\alpha, -\ell_\alpha + 2, \dots, \ell_\alpha$. The matrices A_s use the diagonal parameters D_α in Table 1 and the parameters $P_{\alpha\beta}$ in Table 2, defined only when $\ell_\beta = \ell_\alpha + 2$.

α	1	2	3	4	5	6	7	8
D_α	−289	−84	−9	33	−22	27	71	−11

Table 1: Diagonal parameters, common to both lengths.

Here are complete entry rules. Consider the row (α, d) and column (β, e) , put $\ell = \ell_\alpha$, $k = \ell_\beta$, and fix $s \in \{-1, 0, 1\}$. The entry is zero unless

$$e = d + 2s, \quad |k - \ell| \leq 2, \quad (k = \ell \implies \alpha = \beta). \quad (\text{C.4})$$

(α, β)	$P_{\alpha\beta}$	(α, β)	$P_{\alpha\beta}$	(α, β)	$P_{\alpha\beta}$
(1, 5)	-11	(2, 7)	-49	(4, 6)	9
(1, 6)	-35	(3, 5)	-6	(4, 7)	-32
(1, 7)	-28	(3, 6)	-25	(5, 8)	1
(2, 5)	-23	(3, 7)	-20	(6, 8)	2
(2, 6)	16	(4, 5)	-34	(7, 8)	-16

Table 2: Parameters joining virtual types whose labels ℓ differ by 2.

For an allowed entry define

$$p = \begin{cases} D_\alpha, & k = \ell, \\ P_{\alpha\beta}, & k = \ell + 2, \\ -P_{\beta\alpha}, & k = \ell - 2. \end{cases}$$

Put $r = (d + \ell)/2$ and set the entry equal to pf , where

	$s = -1$	$s = 0$	$s = 1$
$k = \ell + 2$	$2(\ell + 1 - r)(\ell + 2 - r)$	$2(r + 1)(\ell - r + 1)$	$(r + 1)(r + 2)$
$k = \ell$	$2(\ell - r + 1)$	d	$-(r + 1)$
$k = \ell - 2$	2	-2	1

(C.5)

lists the values of f . These entry rules are independent of N .

C.3 Finite contractions and variational bounds

We now evaluate the contractions in Lemma C.1 for this fixed triple at $N = 72$ and $N = 120$. The support rule (C.4) makes this a short block calculation. On paired virtual labels $((\alpha, d), (\beta, e))$, both T and O preserve $d - e$. For T , the two layers shift by the same $2s$. For O , the two layers have the same total physical weight $s + t$, also in each off-diagonal term. Thus the difference blocks are exhaustive and have dimensions

$d - e$	-10	-8	-6	-4	-2	0	2	4	6	8	10
dimension	1	8	32	80	136	162	136	80	32	8	1

(C.6)

They sum to $676 = 26^2$; each block contributes once.

For completeness, the entire integer computation consists of the following operations. Form A_s , B_{st} , T , and O from Tables 1–2 and Equations (C.2), (C.3), (C.4), and (C.5). Restrict T, O to each difference block, obtaining T_δ, O_δ . Compute powers by

$$P_0 = I, \quad P_{2j} = P_j^2 \quad (j \geq 1), \quad P_{2j+1} = P_j^2 T_\delta \quad (j \geq 0).$$

Then accumulate

$$U_N = \sum_\delta \sum_{i,j} (P_{N-2})_{ij} (O_\delta)_{ji}, \quad V_N = \sum_\delta \sum_{i,j} (P_{N-2})_{ij} (T_\delta^2)_{ji}. \quad (C.7)$$

Only arbitrary-precision integer operations are used. For $N = 72$, the required power is 70. For $N = 120$, one may use $T^{118} = T^{64}T^{32}T^{16}T^4T^2$ and $T^{120} = T^{64}T^{32}T^{16}T^8$ within each block.

The resulting exact integer comparisons are

$V_N > 0,$	$-c_N V_N \leq 10^{12} U_N < (-c_N + 1) V_N,$	$\begin{array}{c c} N & c_N \\ \hline 72 & 1401482112601 \\ 120 & 1401482105174 \end{array}$	(C.8)
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The supplied computation files contain the complete matrices and the full integers U_N, V_N , rather than only these shortened intervals. In particular, the files `verification/computations/evidence/trials/certificate72.json` and `verification/computations/evidence/trials/certificate120.json` record the full contractions and positive margins $-700741V_N - 500000U_N$. The common matrices are recorded in `verification/computations/evidence/trials/matrix-data.json`. The program `verification/computations/trials/run_trial.py`, with `--length 72` or `--length 120`, executes the entry rules and block contractions in `verification/computations/trials/original_trial.py` at the selected length. The integer comparisons in (C.8) also follow directly from their full recorded integers.

Lemma C.2. *For the untwisted periodic spin-one Heisenberg Hamiltonian and $a = 700741/500000$,*

$$E_0(72) < -72a, \quad E_0(120) < -120a.$$

Consequently $Z_{72}(b) > 1$ and $Z_{120}(b) > 1$ for every $b > 0$.

Proof. For each row of (C.8), $(-c_N + 1)/10^{12} < -700741/500000$. Thus the corresponding explicit matrices have $V_N > 0$ and $U_N/V_N < -a$. Apply Lemma C.1. $\square$

The support rule puts these trial vectors in total weight zero, but the variational bound is for the minimum of the full physical spectrum. Their only symmetry used in the contraction proof is cyclic site translation. The trial inputs hold at every positive inverse temperature, so each finite initialization can use them at its own temperature.

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